

CHAPTER 1

INTRODUCTION

Mathematicians have always used sets, e.g., the ancient Greek geometers defined a circle as the set of points at a fixed distance r from a fixed point C , its center. But the systematic study of sets began only at the end of the 19th century with the work of the great German mathematician Georg Cantor, who created a rigorous theory of the concept of *completed infinite* by which we can compare infinite sets as to size. For example, let

$\mathbb{N} = \{0, 1, \dots\}$ = the set of natural numbers,

$\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$ = the set of rational integers,

$\mathbb{Q}$ = the set of rational numbers (fractions),

$\mathbb{R}$ = the points of a straight line,

where we also identify $\mathbb{R}$ with the set of real numbers, each point associated with its (positive or negative) coordinate with respect to a fixed origin and direction. Cantor asked if these four sets “have the same (infinite) number of elements”, or if one of them is “more numerous” than the others. Before we make precise and answer this question in the next chapter, we review here some basic, well-known facts about sets and functions, primarily to explain the notation we will be using.

What are sets, anyway? The question is like “what are points”, which Euclid answered with

a point is that which has no parts.

This is not a rigorous mathematical definition, a reduction of the concept of “point” to other concepts which we already understand, but just an intuitive description which suggests that a point is some thing which has no extension in space. Like that of point, the concept of set is fundamental and cannot be reduced to other, simpler concepts. Cantor described it as follows:

By a set we are to understand any collection into a whole of definite and separate objects of our intuition or our thought.

Vague as it is, this description implies two basic properties of sets.

1. Every set A has **elements** or **members**. We write

$x \in A \iff$ the object x is a member of (or belongs to) A .

2. A set is determined by its members, i.e., if A, B are sets, then¹

$$\begin{aligned} A = B &\iff A \text{ and } B \text{ have the same members} & (1-1) \\ &\iff (\forall x)[x \in A \iff x \in B]. \end{aligned}$$

This last is the **Extensionality Property**. For example, the set of students in this class will not change if we all switch places, lie down or move to another classroom; this set is completely determined by *who we are*, not our posture or the places where we happen to be.

Somewhat peculiar is the **empty set** $\emptyset$ which has no members. The extensionality property implies that *there is only one empty set*.

If A and B are sets, we write

$$A \subseteq B \iff (\forall x)[x \in A \implies x \in B],$$

and if $A \subseteq B$, we call A a **subset** of B , so that for every B ,

$$\emptyset \subseteq B, \quad B \subseteq B.$$

A **proper subset** of B is a subset distinct from B ,

$$A \subsetneq B \iff [A \subseteq B \ \& \ A \neq B].$$

From the extensionality property it follows that for all sets A, B ,

$$A = B \iff A \subseteq B \ \& \ B \subseteq A.$$

We have already used several different notations to define specific sets and we need still more, e.g.,

$$A = \{a_1, a_2, \dots, a_n\}$$

is the (finite) set with members the objects $a_1, a_2, \dots, a_n$. If P is a condition which specifies some property of objects, then

$$A = \{x \mid P(x)\}$$

is the set of all objects which satisfy the condition P , so that for all x ,

$$x \in A \iff P(x).$$

For example, if

$$P(x) \iff x \in \mathbb{N} \ \& \ x \text{ is even},$$

then $\{x \mid P(x)\}$ is the set of all even, natural numbers. We use a variant of this notation when we are only interested in “collecting into a whole” members of a given set A which satisfy a certain condition:

$$\{x \in A \mid P(x)\} =_{\text{df}} \{x \mid x \in A \ \& \ P(x)\},$$

¹We will use systematically, as abbreviations, the logical symbols

$\&$: and, $\vee$: or, $\neg$: not, $\implies$: implies, $\iff$: if and only if,

$\forall$: for all, $\exists$: there exists, $\exists!$: there exists exactly one.

The symbols $=_{\text{df}}$ and $\iff_{\text{df}}$ are read “equal by definition” and “equivalent by definition”.

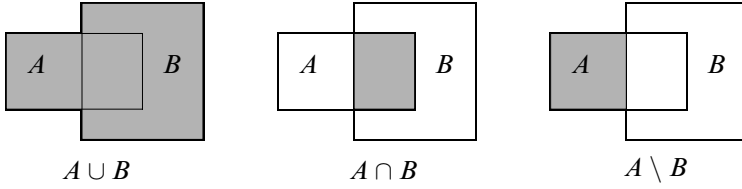


FIGURE 1.1. The Boolean operations.

so that, for example, $\{x \in \mathbb{N} \mid x > 0\}$ is the set of all non-zero natural numbers, while $\{x \in \mathbb{R} \mid x > 0\}$ is the set of all positive real numbers.

For any two sets² A, B ,

$$\begin{aligned} A \cup B &= \{x \mid x \in A \vee x \in B\} && \text{(the union of } A, B), \\ A \cap B &= \{x \in A \mid x \in B\} && \text{(the intersection of } A, B), \\ A \setminus B &= \{x \in A \mid x \notin B\} && \text{(the difference of } A, B). \end{aligned}$$

These “Boolean operations” are illustrated in the so-called *Venn diagrams* of Figure 1.1, in which sets are represented by regions in the plane. The union and the intersection of infinite sequences of sets are defined in the same way,

$$\begin{aligned} \bigcup_{n=0}^{\infty} A_n &= A_0 \cup A_1 \cup \cdots = \{x \mid (\exists n \in \mathbb{N})[x \in A_n]\}, \\ \bigcap_{n=0}^{\infty} A_n &= A_0 \cap A_1 \cap \cdots = \{x \mid (\forall n \in \mathbb{N})[x \in A_n]\}. \end{aligned}$$

Two sets are **disjoint** if their intersection is empty,

$$A \text{ is disjoint from } B \iff A \cap B = \emptyset.$$

We will use the notations

$$f : X \rightarrow Y \text{ or } A \xrightarrow{f} B$$

to indicate that f is a **function** which associates with each member x of the set X , the **domain** of f some member $f(x)$ of the **range** Y of f . Functions are also called **mappings**, **operations**, **transformations** and many other things. Sometimes it is convenient to use the abbreviated notation $(x \mapsto f(x))$ which makes it possible to talk about a function without officially naming it. For example,

$$(x \mapsto x^2 + 1)$$

is the function on the real numbers which assigns to each real its square increased by 1; if we call it f , then it is defined by the formula

$$f(x) = x^2 + 1 \quad (x \in \mathbb{R})$$

²In “mathematical English”, when we say “for any two objects x, y ”, we do not mean that necessarily $x \neq y$, e.g., the assertion that “for any two numbers x, y , $(x+y)^2 = x^2 + 2xy + y^2$ ” implies that “for every number x , $(x+x)^2 = x^2 + 2xx + x^2$ ”.

so that $f(0) = 1$, $f(2) = 5$, etc. But we can say “all the values of $(x \mapsto x^2 + 1)$ are positive reals” without necessarily fixing a name for it, like f .

Two functions are **equal** if they have the same domain and they assign the same value to every member of their common domain,

$$f = g \iff (\forall x \in X)[f(x) = g(x)] \quad (f : X \rightarrow Y, g : X \rightarrow Z, x \in X).$$

In connection with functions we will also use the notations

$$\begin{aligned} f : X \rightarrow Y &\iff_{\text{df}} f \text{ is an } \mathbf{injection} \text{ (one-to-one)} \\ &\iff (\forall x, x' \in X)[f(x) = f(x') \implies x = x'], \end{aligned}$$

$$\begin{aligned} f : X \twoheadrightarrow Y &\iff_{\text{df}} f \text{ is a } \mathbf{surjection} \text{ (onto)} \\ &\iff (\forall y \in Y)(\exists x \in X)[f(x) = y], \end{aligned}$$

$$\begin{aligned} f : X \xrightarrow{\sim} Y &\iff_{\text{df}} f \text{ is a } \mathbf{bijection} \text{ or a } \mathbf{correspondence} \\ &\iff (\forall y \in Y)(\exists! x \in X)[f(x) = y]. \end{aligned}$$

For every $f : X \rightarrow Y$ and $A \subseteq X$, the set

$$f[A] =_{\text{df}} \{f(x) \mid x \in A\}$$

is the **image** of A under f , and if $B \subseteq Y$, then

$$f^{-1}[B] =_{\text{df}} \{x \in X \mid f(x) \in B\}$$

is the **pre-image** of B by f .

If f is a bijection, then we can define the **inverse function** $f^{-1} : Y \rightarrow X$ by the condition

$$f^{-1}(y) = x \iff f(x) = y,$$

and then the inverse image $f^{-1}[B]$ (as we defined it above) is precisely the image of B under f^{-1} .

The **composition**

$$h =_{\text{df}} g \circ f : X \rightarrow Z$$

of two functions

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

is defined by

$$h(x) = g(f(x)) \quad (x \in X).$$

It is easy to prove many basic properties of sets and functions using only these definitions and the extensionality property. For example,

$$A \cup B = B \cup A,$$

because, for any x ,

$$\begin{aligned} x \in A \cup B &\iff x \in A \text{ or } x \in B \\ &\iff x \in B \text{ or } x \in A \\ &\iff x \in B \cup A. \end{aligned}$$

In some cases, the logic of the argument gets a bit complex and it is easier to prove an identity $U = V$ by verifying separately the two implications

$x \in U \implies x \in V$ and $x \in V \implies x \in U$. For example, if $f : X \rightarrow Y$ and $A, B \subseteq X$, then

$$f[A \cup B] = f[A] \cup f[B].$$

To prove this, we show first that

$$x \in f[A \cup B] \implies x \in f[A] \cup f[B];$$

this holds because if $x \in f[A \cup B]$, then there is some $y \in A \cup B$ such that $x = f(y)$; and if $y \in A$, then $x = f(y) \in f[A] \subseteq f[A] \cup f[B]$, while if $y \in B$, then $x = f(y) \in f[B] \subseteq f[A] \cup f[B]$. Next we show the converse implication, that

$$x \in f[A] \cup f[B] \implies x \in f[A \cup B];$$

this holds because if $x \in f[A]$, then $x = f(y)$ for some $y \in A \subseteq A \cup B$, and so $x \in f[A \cup B]$, while if $x \in f[B]$, then $x = f(y)$ for some $y \in B \subseteq A \cup B$, and so, again, $x \in f[A \cup B]$.

Problems for Chapter 1

x1.1. For any three sets A, B, C ,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$$

$$A \setminus (A \cap B) = A \setminus B.$$

x1.2. (De Morgan's laws) For any three sets A, B, C ,

$$C \setminus (A \cup B) = (C \setminus A) \cap (C \setminus B),$$

$$C \setminus (A \cap B) = (C \setminus A) \cup (C \setminus B).$$

x1.3. (De Morgan's laws for sequences) For any set C and any sequence of sets $\{A_n\}_n = A_0, A_1, \dots$,

$$C \setminus (\bigcup_n A_n) = \bigcap_n (C \setminus A_n),$$

$$C \setminus (\bigcap_n A_n) = \bigcup_n (C \setminus A_n).$$

x1.4. For every injection $f : X \rightarrow Y$, and all $A, B \subseteq X$,

$$f[A \cap B] = f[A] \cap f[B],$$

$$f[A \setminus B] = f[A] \setminus f[B].$$

Show also that these identities do not always hold if f is not an injection.

x1.5. For every $f : X \rightarrow Y$, and all $A, B \subseteq Y$,

$$f^{-1}[A \cup B] = f^{-1}[A] \cup f^{-1}[B],$$

$$f^{-1}[A \cap B] = f^{-1}[A] \cap f^{-1}[B],$$

$$f^{-1}[A \setminus B] = f^{-1}[A] \setminus f^{-1}[B].$$

x1.6. For every $f : X \rightarrow Y$ and all sequences of sets $A_n \subseteq X$, $B_n \subseteq Y$,

$$f^{-1}[\bigcup_{n=0}^{\infty} B_n] = \bigcup_{n=0}^{\infty} f^{-1}[B_n],$$

$$f^{-1}[\bigcap_{n=0}^{\infty} B_n] = \bigcap_{n=0}^{\infty} f^{-1}[B_n],$$

$$f[\bigcup_{n=0}^{\infty} A_n] = \bigcup_{n=0}^{\infty} f[A_n].$$

x1.7. For every injection $f : X \rightarrow Y$ and every sequence of sets $A_n \subseteq X$,

$$f[\bigcap_{n=0}^{\infty} A_n] = \bigcap_{n=0}^{\infty} f[A_n].$$

x1.8. The composition of injections is an injection, the composition of surjections is a surjection, and hence the composition of bijections is a bijection.